

Steven Vallejo Ortiz

Software Engineer — Backend · Cloud · AI & agents

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Colombian RUT

PROFESSIONAL PROFILE

Backend software engineer with 13 years of coding —since the game I published on Google Play in 2013— and over a decade of systems in production. Node.js/NestJS, TypeScript, PostgreSQL, Docker, Linux and GCP: I built and run a multi-tenant SaaS with e-commerce, POS with Colombian DIAN e-invoicing and credit management, live with real clients, plus AI automation (LLM, RAG, MCP, OCR) that replaces manual work. I have worked as CTO / Head of Systems, senior backend engineer and SysAdmin. I turn the philosophy I study at Universidad de Antioquia into engineering: executable formal logic with a custom CDCL SAT solver and 6,000+ tests, now running in production agent systems (Cauce V3, Agora).

SKILLS & TECHNOLOGIES (196)

Backend Development: TypeScript · Node.js · NestJS · Hapi · Python · Express · PHP · Laravel · Symfony · Bash · SLIM Framework · .NET · C# · Fastify · FastAPI · Zod (typed contracts) · TypeORM · socket.io (real-time) · Turborepo / monorepos · Vitest · Jest · Testcontainers · esbuild / SWC · better-sqlite3

Frontend Development: ReactJS (Redux, sagas, ReactContext) · Next.js · JavaScript · HTML · CSS3 (Bootstrap, MaterialUI, Bulma) · Flask · PySimpleGUI · Angular · React Native · Laravel Blade / Symfony Twig · Vue.js (VueX) · Tailwind CSS · Vite · Radix UI / shadcn · MDX · xterm.js (web terminal) · D3.js · Phaser 3

DevOps & Cloud: Git, GitHub, BitBucket, GitLab · Firebase · Linux Hosting · Docker · Vercel · Google Cloud · Render · RabbitMQ / Apache Kafka · GitHub Actions · Google Artifacts · AWS · Amazon SQS · Google Pub/Sub · ZeroMQ · Neon · Azure · Docker Compose · Caddy (reverse proxy, TLS) · systemd / cron · Cloud Run · Headscale / Tailscale / NetBird (VPN mesh) · Nginx · ZFS / RAID / NAS · Wazuh (SIEM) · fail2ban / Linux hardening · Debian .deb packaging · Prometheus

Databases: PostgreSQL · MariaDB · MySQL · MongoDB · Neo4j / GraphDB · Neo4j · SQLite · Redis · DynamoDB · Apache Kafka · CouchDB · Cassandra · Elasticsearch · SQL (queries, migrations, indexing, tuning) · SQL Server · ChromaDB (vector store) · Upstash / serverless Redis · Firestore

AI & Machine Learning: Numpy, Wolfram Alpha, NLP · Cupy · TensorFlow, PyTorch · Hugging Face · Stable Diffusion (advanced) · CUDA · Parallelization · Matplotlib · PyTorch · OpenCV · Keras · Pandas · Hybrid RAG (BGE-M3 + BM25) · vLLM · llama.cpp / GGUF (quantization) · faster-whisper (STT) · TTS (Kokoro, ElevenLabs) · OCR (Surya, Tesseract) · Fine-tuning / GPU quantization

AI Tools: ChatGPT · GitHub Copilot / TabNine · OpenAI API · Google Cloud AI · Google Gemini · DeepSeek · Claude (Anthropic) · Llama · Whisper · MCP · Model Context Protocol (own servers) · Claude Code · Codex CLI · Gemini CLI · Ollama · MiniMax · xAI / Grok

Paradigms & Protocols: REST · Object-Oriented · Functional · GraphQL · Procedural · Reactive · Event-Driven · SQS · Logic · Concurrent · WebSocket · RPC · MQTT · Declarative · Aspect-Oriented · SOAP · Structured Programming · Metaprogramming · gRPC

Project Management: SCRUM, KANBAN, LEAN · Advance Documents · Trello, JIRA, ClickUp · TDD, UML · Kanban · Jira · ClickUp · Notion · Excel · Obsidian / living documentation · Runbooks and postmortems

Game Development: C# Unity 3D · Virtual Reality (Oculus, HTC VIVE) · Augmented Reality (Vuforia) · In-game data capture · Roblox · Decentraland · Phaser 3 (2D web)

AI Agents & Orchestration: Agent-fleet orchestration (Cauce V3) · Custom MCP servers · Tool-calling and tool registries · Delivery contracts and multi-tenant routing · Sub-agents and parallel execution · BYOK and model credential management · Multi-LLM with fallback chains · Agent observability and tracing · Evaluation and output guards

Automation & Integrations: RPA / process automation · Chrome DevTools Protocol (CDP) · Puppeteer / Playwright · Google Workspace APIs (Gmail, Calendar, Drive, Docs, Sheets) · Telegram Bot API · WhatsApp Business Cloud API · DIAN e-invoicing (Colombia) · Payment gateways (Wompi, Mercado Pago) · Signed webhooks and idempotent queues · Scraping and structured extraction

Simulation & Computational Methods: Agent-based modelling (ABM) · ODE / PDE (incl. GPU) · Deep reinforcement learning (PyTorch) · CuPy / GPU computing · Permutation tests and ablative validation · SAT solving (CDCL) · Type theory (MLTT) · ffmpeg / programmatic rendering

Philosophy & Logic: Formal logic · Analytic philosophy · Argumentation · Epistemology · Philosophy of mind · Philosophy of AI · Symbolic logic · Ethics · Type theory · SAT solving

PROFESSIONAL EXPERIENCE (FULL HISTORY)

Systems Administrator (SysAdmin) — Critertec

Nov 2024 — Jun 2026 (ended) · Fixed-term contract · Ended 2026 · Medellín, Antioquia, Colombia

Operated and secured the production infrastructure (Linux, Docker, CI/CD) running the agency's digital solutions to ensure continuous service availability. Notable project: Soy Digital / INDOTEL (see below).

Technologies: Linux · Docker · CI/CD · PostgreSQL · SQL · server administration · backups · hardening

Frontend Developer / Software Engineering Contractor — Soy Digital / INDOTEL (proyecto de gobierno, República Dominicana)

Nov 2024 — Jun 2026 (~2 years) · via Critertec · Service contract · Government project · República Dominicana (remoto)

Built the frontend with offline support and automatic sync for a national digital-skills platform of the Dominican government (INDOTEL), using React/Next.js PWA so registration worked without a stable connection in the field. Data analysis and audit dashboards. Software engineering contractor at Critertec for Soy Digital, an INDOTEL initiative in the Dominican Republic.

Technologies: React · Next.js · PWA · offline-first · automatic sync · SQL · audit dashboards

Founder / Lead Developer — Humanizar Systems

2022 — present · Own project (venture) · In production · Medellín, Antioquia, Colombia

Built an enterprise multi-tenant SaaS ecosystem —e-commerce (Graf), POS with DIAN electronic invoicing (Sinergia POS), credit fintech (FIAR), and WhatsApp messaging (EMW)— using NestJS microservices with pub/sub communication and HMAC-signed webhooks, JWT/OAuth2 auth and CI/CD deployment, so SMBs could sell, invoice legally and manage credit from a single platform. Running in production with real clients.

Technologies: Node.js · NestJS · TypeScript · PostgreSQL · SQL · Docker · microservices · pub/sub · HMAC webhooks · JWT/OAuth2 · CI/CD · React · Next.js · DIAN e-invoicing · Wompi · Mercado Pago

CTO / Systems Manager — Finca Directa S.A.S

Nov 2021 — present · Service contract · Active · Medellín, Antioquia, Colombia

Lead production systems and built a real enterprise automation harness (RPA): email classification, bank-statement downloads, invoice OCR, CRM sync — with idempotency, human-in-the-loop validation and a decision log. Eliminated day-to-day manual repetitive work.

Technologies: Node.js · Python · SQL · PostgreSQL · Linux · OCR · RPA · CRM integration · idempotency · webhooks

Backend Developer — Indie Levels Studio S.A.S

Aug 2021 — present · Fixed-term contract · Active · Medellín, Antioquia, Colombia

Developed and maintained backend REST APIs (Node.js/TypeScript + PostgreSQL) for a video game studio's products used by clients worldwide, with Git version control and testing to keep releases stable.

Technologies: Node.js · TypeScript · REST APIs · PostgreSQL · SQL · Git · automated testing

Semi-Senior Full Stack Developer — ZENIT S.A.S

Oct 2020 — Aug 2021 · Employment · Medellín, Antioquia, Colombia

Junior Full Stack Developer — INS S.A.S

Feb 2019 — Oct 2020 · Employment · Medellín, Antioquia, Colombia

Junior Full Stack Developer — Kambban S.A.S

Feb 2020 — May 2020 · Employment · Medellín, Antioquia, Colombia

Intern at SIPAEM research group — SENA / SIPAEM

Mar 2018 — Oct 2018 · Internship · Medellín, Antioquia, Colombia

Manager - Developer — IQpixels S.A.S

Jul 2016 — Jan 2017 · Employment · Medellín, Antioquia, Colombia

Freelance — IT Infrastructure — Independiente

Feb 2014 — present · Freelance · Medellín, Antioquia, Colombia

Freelance in IT systems infrastructure.

Technologies: Linux · Docker · networking · hosting · backups · virtualization · monitoring

Freelance — Video Game Development — Independiente

2013 — present · Freelance · Medellín, Antioquia, Colombia

Independent video game development since 2013, when I published my first game on Google Play. It is the starting point of my career in programming.

Technologies: Unity 3D · C# · Roblox · virtual reality (Oculus, HTC VIVE) · augmented reality (Vuforia)

Freelance — Web App Development — Independiente

2018 — present · Freelance · Medellín, Antioquia, Colombia

Freelance in web application development.

Technologies: React · Next.js · Node.js · PHP · Laravel · MySQL / MariaDB · SQL

MAIN OWN PROJECTS

Cauce V3 · <https://humanizar.tech>

Built the orchestrator for the Humanizar agent fleet: deployments, monitoring, delivery contracts between agents and a multi-tenant CRM. My flagship engineering project in Mouseñon.

Stack: Orquestación agéntica · Contratos de entrega · Multi-tenant

Ágora — Plataforma académica y agentes IA · <https://agora.elenxos.com> ·

<https://github.com/stevenvo780/EducacionCooperativa> · <https://github.com/stevenvo780/agora-backend>

Built an AI-agent SaaS platform using a poly-repo of separate services (Next.js frontend, Cloud Run backend, a real-time socket.io hub and Docker workers) to isolate concerns and deploy each part independently. Design decisions: typed contracts validated with Zod, HMAC-signed webhooks, secrets encrypted with AES-256-GCM, and a custom MCP server. I designed and implemented the architecture, backend, frontend and DevOps.

Stack: Next.js 15 · React 18 · Redux · Express/TypeScript (Cloud Run) · socket.io · Docker PTY · Zod · HMAC · MCP server

Ecosistema Humanizar · <https://www.humanizar.co/>

Multi-tenant enterprise SaaS ecosystem I built: e-commerce (Graf), point-of-sale with DIAN e-invoicing (Sinergia POS), credit fintech (FIAR), and WhatsApp messaging (EMW), orchestrated with NestJS microservices, pub/sub and HMAC-signed webhooks.

Stack: NestJS microservices · TypeORM · PostgreSQL · MySQL · Redis · Next.js · React 18 PWA · pub/sub · HMAC webhooks

ST — Lenguaje de lógica formal ejecutable · <https://github.com/stevenvo780/ST> · <https://organon.stevenvallejo.com> ·

<https://www.npmjs.com/package/@stevenvo780/st-lang>

Executable formal-logic language (npm `@stevenvo780/st-lang`): a DSL with 30+ logic profiles, a custom SAT solver (CDCL), type theory (MLTT), and 6,333+ tests. The most ambitious project in the portfolio; integrated into the Agora agent. A bridge between analytic philosophy and engineering.

Stack: TypeScript · SAT solver CDCL · MLTT (teoría de tipos) · 30+ perfiles lógicos · bilingüe

auto.logic · <https://github.com/stevenvo780/auto.logic> · <https://www.npmjs.com/package/@stevenvo780/autologic>

Natural language to executable formal logic with no AI, using rule-based NLP only: 11 logic profiles, ~200 discourse markers, zero external deps, bilingual. A natural companion to ST.

Stack: TypeScript · NLP basado en reglas · 11 perfiles lógicos · ~200 marcadores discursivos · zero deps

Estructuras Pre-Ontológicas — motor de simulación científica · <https://estructuras-preontologicas.vercel.app> ·

<https://github.com/stevenvo780/EstructurasPreontologicas>

Computational engineering for a doctoral research project in complexity science. I built the simulation engine coupling agent-based models (ABM) and differential equations (ODE), plus the statistical validation pipeline via ablative intervention: permutation tests (999), bootstrap (500) and AUC-ROC against an ARIMA baseline. It includes a hostile-testing harness (0/2000 false positives under random walk) and simulations spanning 30 orders of magnitude in space and time with physically-motivated probes. Custom CLI, isolated environment, CPU/GPU.

Stack: Python · numpy/scipy/pandas · ABM+ODE · PyTorch/CUDA · permutación/bootstrap · AUC-ROC · CLI

PORTFOLIO

Programming: Symploque CRM SaaS · ST — Lenguaje lógica formal · Agora (plataforma) · MCP Delegate Agents · auto.logic · Agora (frontend) — EducacionCooperativa · Agora (backend) · Helenikos · Graf (e-commerce) · Sinergia POS · FIAR (frontend) · Ultimate Terminal · Cristina CMS · Cafetería del Caos (Next) · Jarvis IA V2 · Chat LLM Local (GPU) · Warehouse Management · EMW · PDF to Markdown IA · MCP Agents · FIAR API · Cafetería del Caos API · Cafetería del Caos Frontend · Cafetería del Caos Bot · Debates Util · Template React Big APP · Aurora CQRS · API Node MongoDB · GraphQL Example · Laravel React REST · Neutrino Sugar · Ethical Hacking Tool · Mi CV

Literature & Philosophy: Estructuras Pre-Ontológicas · Tesis Jacob — Motor de Hiperobjetos · Fenomenología Urbana · Motor de vida artificial (ABM + RAG) · Redes Neuronales, Filosofía y Neurociencia · Mundo simulado (ABM + PDE + RAG) · Simulación con planificación por objetivos · Simulaciones de Hiperobjetos · Logic Education · Dúo Eterno · Artículos de Filosofía (Medium)

Mathematics & Sciences: Algoritmo de Entrenamiento del Universo · Fenómenos Emergentes — Juego de la Vida de Conway · Modelo MASOES (Python) · Teoría de la Complejidad · Sistema Económico Simulado · Teoría de la Información en Comunicación Celular · Teoría de Juegos · Teoría del Caos y Atractor de Lorenz · Teoría de la Decisión · Gráfico RULIAT Extendido · Sistema de Transporte en Trenes · Experimento de Temperatura Emergente

Games: Cash: El Resurgir de los Imperios · Escape Room ILS · ILS TEST (Roblox Game)

Other notable repos: tertulialiteraria-api · tertulialiteraria-frond · SongArt (mauroLetraCancionesImágenes) · jarvisIA · kratos-jarvis

EDUCATION · LANGUAGES

Philosophy (degree) — Universidad de Antioquia · 2025 — est. 2029 · Active student

Software Development Technician — SENA · 2017 · Graduate (formal diploma)

HTC VR Development — SENA · 2017 · Completed

Video Game Development — CENSA · 2017 · Completed

Languages: Spanish — Native · English — Strong technical reading and writing; conversational at a working level (B1).